

the GNU Emacs manual at https://www.gnu.org/software/emacs/manual/html_node/emacs/Calendar_002fDiary.html#Calendar_002fDiary, to learn more.

Centering

Let's assume you've written a poem where both the verses and the title appear left-margin aligned. So you want to centre the title and the text. You can do so by first placing the cursor on the title/text, and typing 'M-x set-justification-center'.

Marking and highlighting the poem, and using 'M-x center-region' will center the poem.

Macros

Macros are recorded key strokes that can be stored and replayed. You can start defining a keyboard macro using the 'C-x (' command or the F3 key. You can then type a series of keys that constitute the macro. To finish defining the macro, you can use 'C-x)' or the F4 key. In order to execute the previously defined macro, you can use the 'C-x e' shortcut or F4.

Consider the following text in a buffer that contains a serial number, date and an examination subject list:

```
1,2015-03-02,English
2,2015-03-03,Physics
3,2015-03-05,Mathematics
4,2015-03-08,Biology
5,2015-03-10,Chemistry
```

Suppose you wish to add a space after each comma, you can define the following macro (exclude the semi-colon followed by the text) for the first line using the following key strokes:

```
F3 ; Start macro definition
C-s ; Search for
, ; comma
Enter ;
Space ;
C-s ; Search again for
, ; comma
Enter ;
Space ;
C-n ; Move to next line
C-a ; Move to beginning of line
F4 ; End macro definition
```

Using 'C-x e' or F4 repeatedly will turn the above input CSV text into the following:

```
1, 2015-03-02, English
2, 2015-03-03, Physics
3, 2015-03-05, Mathematics
4, 2015-03-08, Biology
5, 2015-03-10, Chemistry
```

You can give a name (say, 'comma') to the previously

defined macro using 'C-x C-k n'. You can then execute the macro using 'M-x comma'. You can also insert the named macro into a file using the 'M-x insert-kbd-macro' command. You can bind a macro to a key using the 'C-x C-k b' shortcut.

If you wish to apply the macro to each line in a region, you can use 'C-x C-k r' keys. In order to cycle between the previous and next macros in the macro ring, you can use the 'C-x C-k C-p' and 'C-x C-k C-n' shortcuts respectively. You can also delete a macro using the 'C-x C-k C-d' key combination.

Picture mode

You can draw diagrams inside Emacs using *Picture mode*. To start *Picture mode*, use the 'M-x picture-mode' command, and to exit, use the 'C-c C-c' shortcut.

The cursor movement keys in a buffer are also applicable in *Picture mode*. To move the cursor right, you can use the 'C-f' keys, and to move left by one character, you can use the 'C-b' shortcut. To move the cursor up and down by one character, use the 'C-p' and 'C-n' shortcuts, respectively. The 'C-d' shortcut is used to delete a character.

Before you move the cursor to draw in the buffer, you need to set the drawing direction. Table 1 summarises the shortcut keys and their associated drawing direction.

Table 1

Shortcut	Direction
C-c Ctrl	Up
C-c .	Down
C-c >	Right
C-c <	Left
C-c ^	Northwest
C-c ' (apostrophe)	Northeast
c /	Southwest
C-c \	Southeast

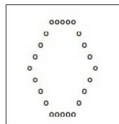


Figure 2: Diagram using *Picture mode*

If you want to move the cursor forward in the drawing direction, you can use the 'C-c C-f' shortcut. To move the cursor backward, use the 'C-c C-b' key combination. If you want to delete a line, use the 'C-k' command. You can insert a new line using the 'C-o' shortcut. You can also draw a rectangle around a region using the 'C-c C-r' shortcut. A drawing done using *Picture mode* is shown in Figure 2.

Artist mode

Artist mode can also be used to draw diagrams in GNU Emacs. You can enter this mode using 'M-x artist-mode' and exit the same using 'C-c C-c'.

You can draw pictures using the keyboard alone or also use the mouse in *Artist mode*. In order to start and stop drawing, use the *Enter* key. This is equivalent to putting the